



Enhancing Alzheimer's Disease Classification using Transformer based Neuroimaging Technique

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Abstract

As a major global health challenge, the progressive cognitive decline and memory loss associated with Alzheimer's disease (AD) significantly impact individuals' quality of life. Due to the lack of a definitive cure, early and precise diagnosis remains crucial for implementing effective intervention and management strategies. In this study, we introduce a pioneering approach utilizing Transformer based ResLadderNet for AD classification, achieving an exceptional accuracy of 98%. The classification of AD is crucial for several reasons. Firstly, accurate classification aids in early detection, allowing healthcare professionals to implement interventions aimed at mitigating disease progression and enhancing patient outcomes. Secondly, it allows for stratification of patients based on disease severity and subtype, which is essential for personalized treatment planning and clinical trials. Moreover, effective classification supports research efforts aimed at understanding disease mechanisms and identifying biomarkers that could further enhance diagnostic accuracy and therapeutic development. The ResLadderNet architecture leverages the strengths of residual networks and ladder networks, optimizing feature extraction and hierarchical representation learning with transformer to feature propagation. This approach not only enhances the interpretability of the model's decisions but also improves its robustness against variations in imaging data quality and disease progression stages. Our methodology employs a meticulously curated dataset of 10,240 samples across four distinct classes, sourced from Harvard, to train and validate the model. This model achieves outstanding performance with an accuracy of 98.93%. Notably, this study demonstrates remarkable performance, achieving precision at 0.98, recall at 0.97, and an F1 score of 0.98, underscoring its efficacy in detecting early indicators of Alzheimer's disease. This breakthrough has significant implications for early intervention and treatment, potentially improving the lives of millions worldwide.

Keywords Alzheimer's Disease · ResLadderNet · Ensemble learning · ViT · CNN · Feature extraction · Feature propagation

Introduction

Alzheimer's Disease (AD) accounts for a substantial portion of dementia cases globally and is characterized as a progressive neurological disorder. It typically starts with mild symptoms, including memory loss and changes in behavior, and gradually advances to profound cognitive decline and impaired functionality. AD affects millions worldwide, with projections indicating a significant increase in cases in the coming decades. The disease progresses through stages from mild memory loss and confusion to profound cognitive decline, often resulting in dependence on caregivers for daily activities. Early diagnosis is crucial for timely interventions, but current methods are often invasive and costly, driving exploration of alternative approaches. Recent advancements in digital imaging and deep learning [1] show promise in improving AD detection accuracy. Integrating techniques

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like Magnetic Resonance Imaging (MRI) and Positron Emission Tomography (PET) with deep learning algorithms enhances the identification of potential AD biomarkers.

The hippocampus, crucial for memory and spatial navigation, is one of the earliest regions affected by AD, underscoring the importance of early intervention. Other affected brain structures include the thalamus, hypothalamus, amygdala, and basal ganglia, contributing to motor and sensory difficulties. While AD remains incurable, treatments aimed at slowing its progression exist, and early diagnosis can significantly enhance patients' quality of life. Risk factors such as age, genetics, and lifestyle influence disease development. Diagnostic procedures include cognitive assessments, neuropsychological evaluations, and non-invasive brain imaging using tools like the Mini-Mental State Examination (MMSE), [2] Montreal Cognitive Assessment (MoCA), MRI, and PET scans. MRI generates detailed brain images using radio waves and magnetic fields, while PET detects brain function changes with radioactive substances, providing critical diagnostic insights without invasive procedures.

While DL-based AD x-ray image classification has advanced, challenges persist, including inconsistent accuracy across varied datasets and difficulties in early-stage detection. Many existing models face limitations in multi-class classification [3] and generalization to new data. Additionally, CNN architectures like ResNet50 and LadderNet, though effective, may not fully capture complex spatial patterns in brain imaging. To address these issues, our study proposes the Trans ResLadderNet model, integrating ResNet50, LadderNet, and transformer-based techniques to improve feature extraction and classification accuracy across four AD progression classes.

Our study leverages these advancements with a dataset of 20,480 images, covering four AD progression classes — No Impairment, Very Mild Impairment, Mild Impairment, and Moderate Impairment — achieved by initial collection of 10,240 real images followed by augmentation and GAN-based synthesis to double the dataset size. Using ResNet 50, LadderNet, and our novel Trans ResLadderNet algorithms, we achieve significant advancements in AD classification accuracy. Trans ResLadderNet, in particular, demonstrates impressive train accuracy of 98% and test accuracy of 97%, supported by high precision of 0.98%, recall value of 0.97, and F1_score (0.98). To improve model transparency and interpretability, we applied Gradient-weighted Class Activation Mapping (Grad-CAM) to visualize important regions influencing classification decisions. Beyond enhancing algorithmic metrics, our meticulous dataset curation ensures robust model training and validation, contributing to the broader AD research community with a benchmark dataset for future studies in AD classification and diagnosis. These insights pave the way for improved diagnostic accuracy, personalized treatment strategies, and advancements

in neurodegenerative disease research, ultimately aiming to improve outcomes for AD patients worldwide.

Literature Review

Duaa A. et al. [4] proposes using a fine tuned CNN deep learning model, and ResNet50, for automatic feature extraction in Alzheimer's diagnosis from MRI images. The results show high accuracy (85.7–99%) outperforming other state-of-the-art models, bold the potential of Deep learning in improving diagnosis and patient outcomes. M. Neethu et al. [5] proposed a novel MCLSO-SpinalNet model for Alzheimer's disease classification, leveraging adaptive median filtering, K-Net segmentation, and feature extraction. The model achieves high accuracy (92.6%), sensitivity (93.6%), F-measure (91.7%), and specificity (93.3%), outperforming existing methods and showcasing its potential for accurate AD diagnosis. Yousry A. et al. [6] proposed a CNN-based framework for same classification achieves high accuracy (99.6–99.8% for binary and 97.5% for multi-class classification) on the ADNI dataset, outperforming existing methods and demonstrating the potential for accurate early diagnosis and improved patient outcomes.

Bharath M. et al. [7] examined various machine learning techniques for early Alzheimer's detection using 6,400 pre-processed MRI images. The SVM model with a linear kernel achieved 98% accuracy for detection and 98.7% for classification, demonstrating its precision in AD diagnosis. Waleed A. S. [8] introduced a deep learning model combining DenseNet-169 and ResNet-50 for diagnosing and classifying Alzheimer's disease. DenseNet-169 showed superior performance over ResNet-50, with training and testing accuracies of 0.977 and 0.8382, respectively, supporting real-time analysis across four disease stages. Ahmad W. S. et al. [9] utilized CNNs with MRI data to identify Alzheimer's disease, achieving a 99% accuracy rate. This deep learning approach outperformed conventional machine learning methods, proving efficient in managing extensive medical datasets and supporting early diagnosis. Farhana I. et al. [10] proposes a multi-stage Alzheimer's disease classification approach, leveraging pre-trained CNNs for feature extraction, principal component analysis for optimization, and conventional machine learning classifiers, achieving 98.71% accuracy on the ADNI dataset with SVM and ResNet50 features. Bo Liu [11] presents a novel classification method, incorporating hippocampus MRI images and a pre-trained DenseNet with attention mechanism (DenseNet-AM). The approach achieves an accuracy of 92.8% in cognitive normal and AD classification, offering promising assistance for doctors in disease diagnosis.

Pengxu L et al. [12] addresses classification challenges between AD and Mild Cognitive Impairment (MCI) using a

multi-classification method. The ConvNeXt network extracts features, and Dynamic Multilayer Perceptron (DMLP) enhances them. The method achieves higher accuracy (78.95%) and F1-score (77.08%) than classical methods, enabling deeper understanding and diagnosis of AD's progressive stages. Rukesh P. et al. [13] proposed a deep neural network incorporating fully connected layers and optimized activation functions (Leaky ReLU, ELU, PReLU), achieving notable accuracy in differentiating Alzheimer's disease (AD) from cognitively normal (CN) subjects. The model demonstrated superior performance compared to conventional machine learning methods, with accuracy rates of 85.19% for AD vs. CN, 76.93% for MCI vs. CN, and 72.73% for AD vs. MCI. Nitika G. et al. [14] implemented a Three-dimensional Ensemble Net utilizing AV45 PET scans and 3D patch extraction, achieving an accuracy of 92.13%, marking it as one of the top-performing models for AD classification. Latifa H. et al. [15] integrated two-dimensional deep convolutional neural networks (CNN) with support vector machines (SVM), achieving exceptional performance, with accuracy rates of 99.79% for AD/CN, 99.6% for AD/MCI, and 97.00% for MCI/CN. Similarly, Marwa Geneedy et al. [16] developed a deep learning model that utilizes a shallow CNN in combination with 2D T1-weighted MRI data, reaching an accuracy of 99.68%. The robustness and high accuracy of their model are pivotal for rapid and dependable AD diagnosis, addressing both global (distinguishing normal, MCI, and AD) and local classification challenges. R.R. J. et al. [17] used a deep learning approach combining fMRI and PET images with a CNN (VGG-16) model, achieving 99.95% accuracy with fMRI and 73.46% with PET images.

Deep learning models, especially CNN-based architectures like ResNet50, DenseNet, and VGG-16, have shown high accuracy (85.7–99.95%) in Alzheimer's disease classification using MRI and PET data. Innovative hybrid methods, such as combining CNN with SVM or integrating attention mechanisms, as well as new architectures like SpinalNet and ConvNeXt, have enhanced diagnostic performance. However, challenges persist in multi-class classification, detecting early AD stages, and ensuring model robustness across various datasets. Exploring transformer integration with CNNs could address these limitations.

Research Design

Our research is centered on the application of ResNet50 and LadderNet models for image classification. These models were selected based on their effective feature extraction capabilities, specifically tailored for accurate classification tasks such as Alzheimer's Disease detection and progression assessment.

Architecture of Resnet50

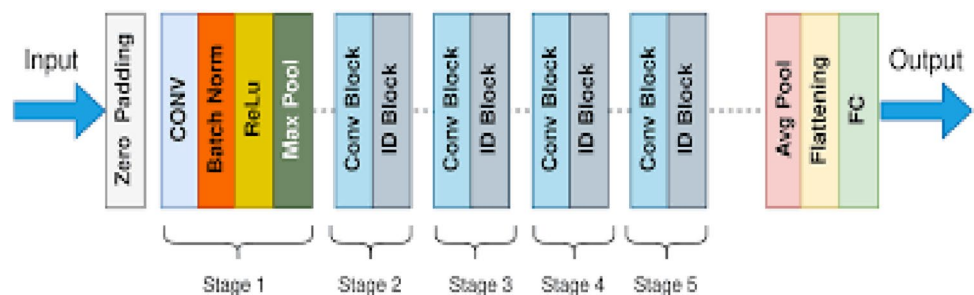
ResNet50 [18] is a prominent deep learning architecture known for its capability to extract complex features from images with high precision. Its 50-layer design incorporates residual connections, skip connections, which address the challenge of vanishing gradients encountered in deep network training by enabling more direct gradient flow during back-propagation. Each ResNet50 block includes three convolutional layers within a bottleneck structure, optimizing computational efficiency while enhancing the representation of features as shown in Fig. 1.

Unlike conventional architectures, ResNet50 employs global average pooling to condense feature maps by calculating their average values, thereby reducing spatial dimensions while preserving crucial features. This approach culminates in a softmax layer that outputs class probabilities, making ResNet50 particularly adept for tasks demanding robust and high-dimensional feature extraction, such as medical imaging analysis and disease diagnosis.

Architecture of LadderNet

LadderNet is a specialized deep learning architecture [19] designed specifically for semantic segmentation tasks, particularly in fields like medical imaging. It combines an encoder-decoder structure with lateral and skip connections to enhance feature propagation and preserve spatial information effectively. Starting with the encoder, LadderNet employs convolutional layers and pooling operations to extract features while progressively reducing spatial dimensions. Meanwhile, the decoder utilizes upsampling

Fig. 1 Architecture of Resnet50



techniques and convolutional layers to reconstruct spatial details and refine feature representations. In its architecture, lateral connections connect matching layers from the encoder to the decoder, promoting the transmission of high-resolution information and gradients, thereby preserving spatial context throughout the network, as illustrated in Fig. 2.

Additionally, skip connections merge features from early layers directly into deeper layers, aiding in the recovery of fine details lost during downsampling. The architecture culminates in a softmax output layer, producing pixel-wise probability maps that accurately delineate distinct classes or segments, proving invaluable for precise segmentation tasks in medical imaging and other complex domains.

Methodology

Flow of Work

The proposed framework begins with data preparation, including normalization and image preprocessing using augmentation, GAN, and SOTA techniques. Feature extraction follows, after which the Transformer-based ResLadderNet model is applied for Alzheimer's classification as shown in Fig. 3. The model performance is validated using sevenfold cross-validation.

Data Collection

The dataset used in this study was obtained from an open-source database provided by Harvard, accessible at

Harvard Medical School AANLIB, which contain 10,240 number of image with 4 different classes including Mild Impairment, Moderate Impairment, No Impairment, and Very Mild Impairment each classes contain 2560 number of images. The sample images of collected one are shown in below Fig. 4.

Data Preprocessing

The subsequent phase focuses on data preprocessing, a crucial step to transform raw data into a format suitable for analysis and training hybrid models. This stage includes several tasks. Normalization adjusts pixel values to a standard scale, typically between 0 and 1, ensuring consistency across the dataset. All images are resized to same dimensions, crucial for batch processing and maintaining consistent model input. To enhance dataset diversity and improve model robustness, basic transformations like rotation, flipping, or cropping are applied for augmentation. These simulate real-world variations, helping the model generalize better. Lastly, noise reduction techniques are used to eliminate extra noise from images, enabling the model and enhancing clarity to learn meaningful patterns effectively.

Data Augmentation

To enhance the robustness and generalization of proposed AD classification model, we apply three data augmentation techniques to our collected dataset of AD images.

a. Rotation and Flipping

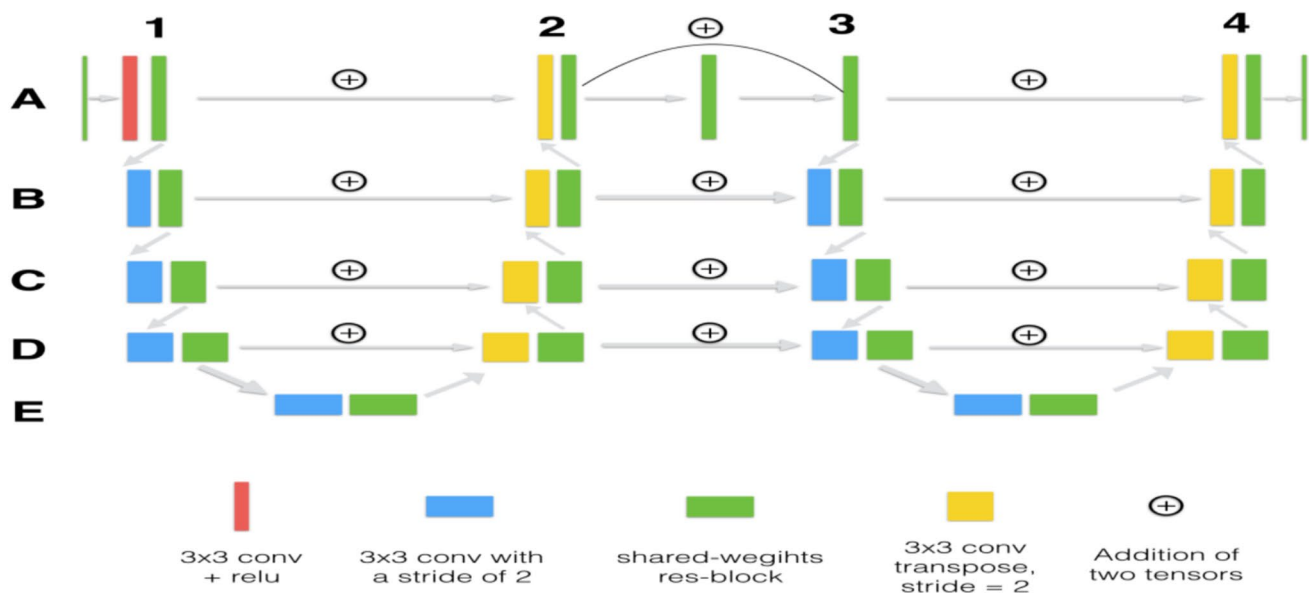


Fig. 2 Architecture of Ladder Net

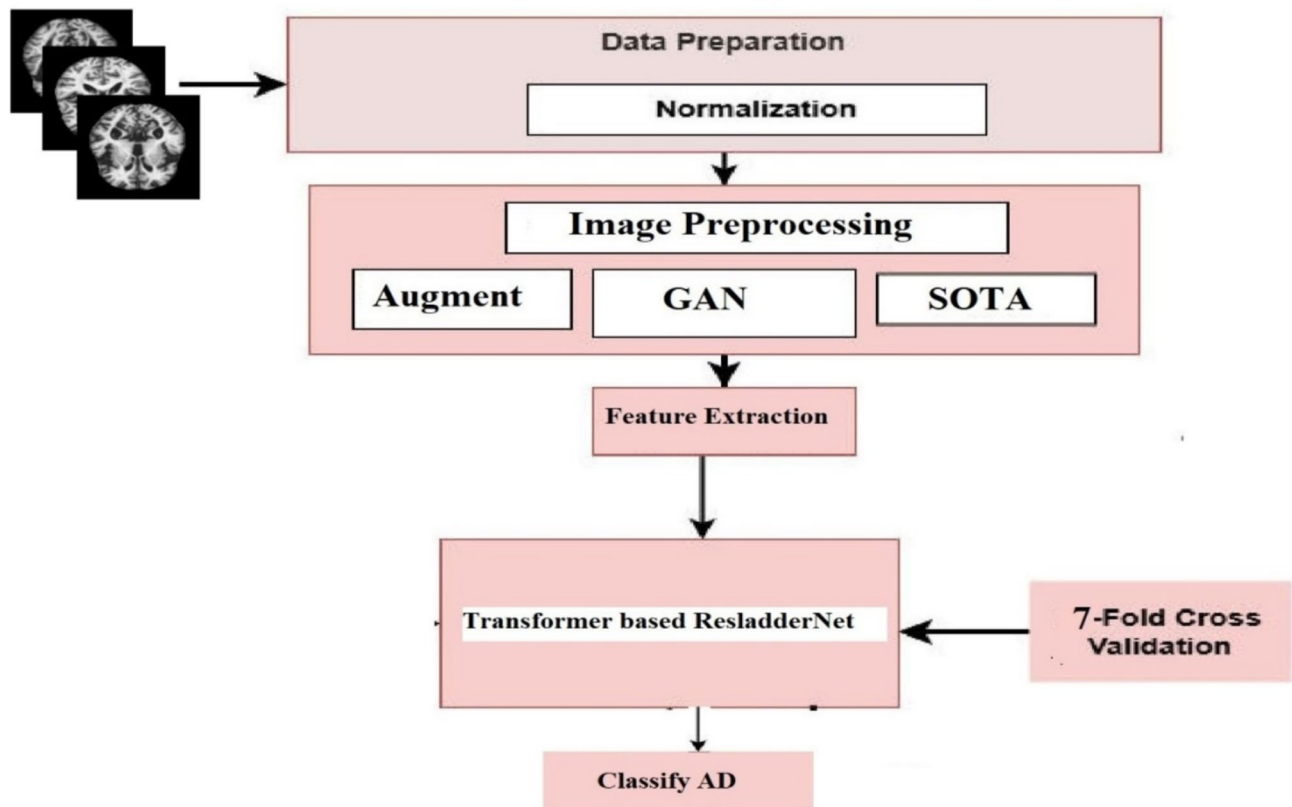


Fig. 3 Work flow of proposed work

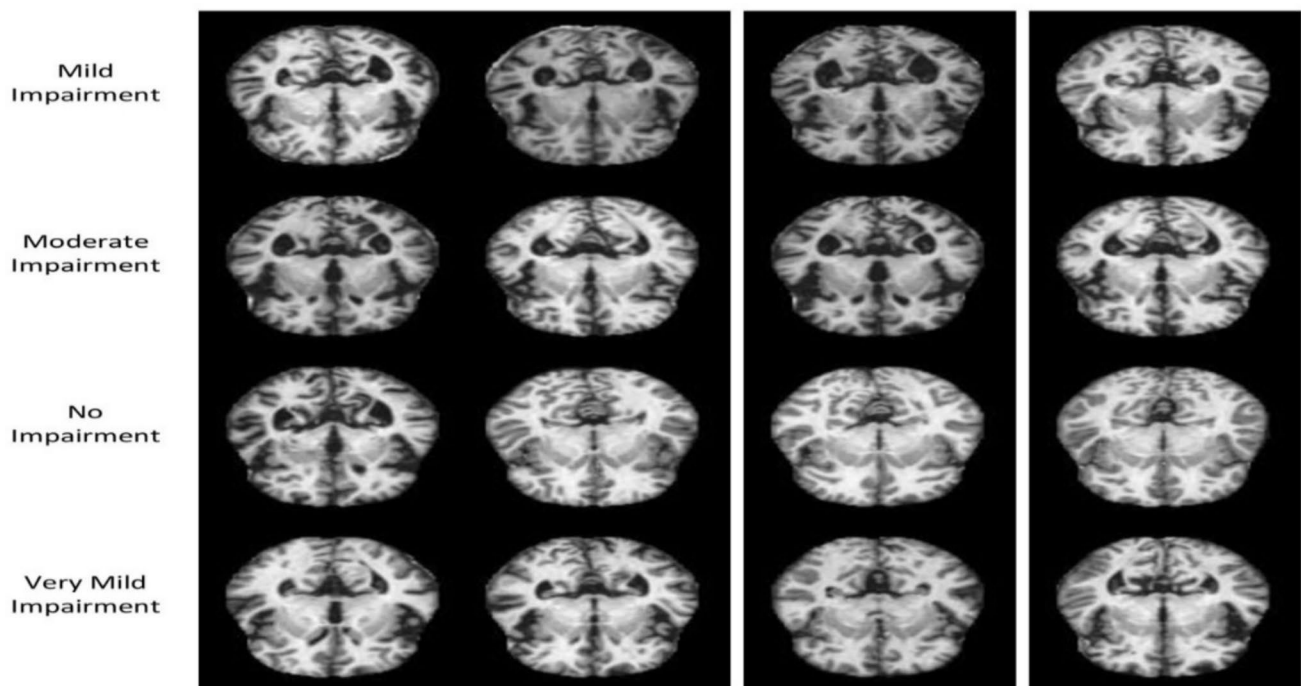


Fig. 4 Sample images of AD

- b. Random Crop
- c. Brightness and Contrast Adjustment

These augmentation techniques help in expanding the diversity of the dataset used for training. By applying them, the model becomes more resilient to variations it might encounter during real-world application scenarios, thus improving its ability to generalize from the training data to unseen instances. Each technique can be implemented with careful consideration of the specific characteristics and requirements of the Alzheimer's Disease image dataset.

Synthetic Image Generate Using GAN

To enhance the dataset and prevent overfitting, we employed a Generative Adversarial Network (GAN) [20] to create synthetic images. This approach allowed us to expand the dataset while preserving diversity. Initially, the dataset contained 10,240 images, but after applying State-of-the-Art (SOTA) augmentation techniques, its size doubled. Synthetic formation sample image is shown in below Fig. 5. This increase in data volume helps the model generalize better, ensuring improved feature learning and reducing the chances of overfitting during classification.

ResLadderNet

We propose ResLadderNet, an innovative deep learning model that merges the strengths of ResNet50 and LadderNet architectures for image classification. ResNet50 [21] is utilized for its robust feature extraction, leveraging

deep convolutional layers to effectively capture intricate patterns from used images. These chosen features are then processed by LadderNet, originally designed for semantic segmentation but adapted here for classification. LadderNet enhances classification accuracy by preserving spatial information via lateral connections and integrating high-resolution details through skip connections. ResLadderNet is designed to excel in image classification tasks, particularly in complex fields like medical imaging and disease diagnosis, where precise feature extraction and classification are crucial. This hybrid model harnesses ResNet50's deep feature learning capabilities while integrating LadderNet's specialized architecture tailored for detailed classification tasks. ResLadderNet represents a significant advancement leveraging combined strengths to increase accuracy and performance in image analysis, promising new opportunities for application in biomedical research and diagnostic settings (Table 1).

ResNet50 is employed for its powerful feature extraction capabilities from MRI scans, utilizing its pre-trained convolutional layers to identify critical patterns related to Alzheimer's disease. LadderNet is then used as encoder-decoder architecture, further refining these extracted features for comprehensive analysis. This combination forms a hybrid model that enhances the classification accuracy of AD into different stages. The integration of ResNet50 and LadderNet [22] enables the model to detect subtle brain changes, facilitating early diagnosis and potentially improving patient outcomes through timely intervention. This approach offers a best solution for non-invasive, accurate Alzheimer's detection. Algorithm of our proposed model is given below.

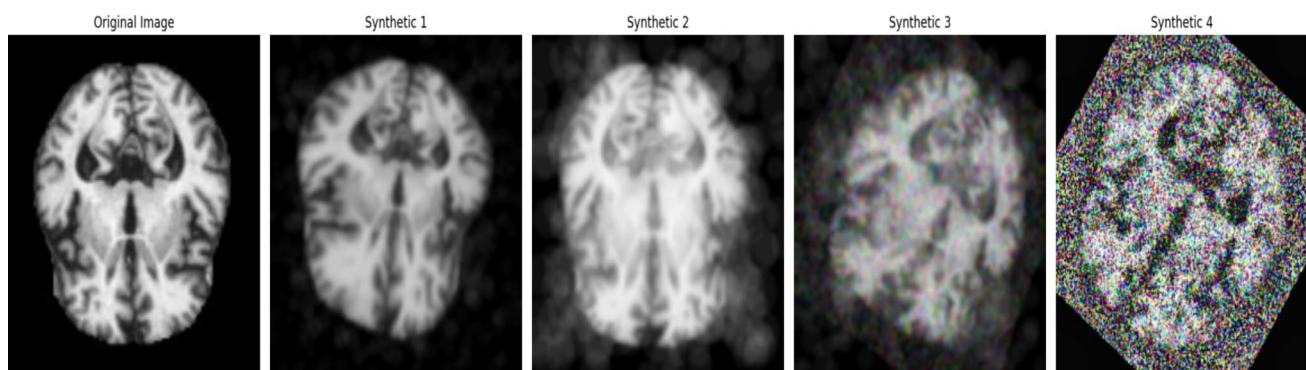


Fig. 5 Formation of synthetic sample

Table 1 Architecture of proposed model

Layer	Type	Kernel Size/Units	Filters/Units	Stride	Activation/Other
Input Layer	-	(224, 224, 3)	-	-	-
ResNet50 Feature Extractor	<i>(Included in the pre-trained ImageNet_top = False)</i>				
Conv2D	7 × 7	64	2	Padding = 'same'	
MaxPooling2D	3 × 3	-	2	-	
Conv_2D	1 × 1	64	1	-	
Conv_2D	3 × 3	64	1	Padding = 'same'	
Conv2D	1 × 1	256	1	-	
MaxPooling2D	3 × 3	-	2	-	
Conv_2D	1 × 1	128	1	-	
Conv_2D	3 × 3	128	1	Padding = 'same'	
Conv_2D	1 × 1	512	1	-	
MaxPooling2D	3 × 3	-	2	-	
Conv2D	1 × 1	256	1	-	
Conv2D	3 × 3	256	1	Padding = 'same'	
Conv2D	1 × 1	1024	1	-	
MaxPooling2D	3 × 3	-	2	-	
Conv2D	1 × 1	512	1	-	
Conv2D	3 × 3	512	1	Padding = 'same'	
Conv2D	1 × 1	2048	1	-	
Encoder Part of LadderNet	Conv2D	3 × 3	512	-	Activation = 'Relu', Padding = 'Rame'
	MaxPooling_2D	2 × 2	-	Pool size = 2	-
	Dropout	-	-	-	0.5
Decoder Part of LadderNet	UpSampling_2D	2 × 2	-	-	-
	Conv_2D	3 × 3	512	-	Activation = 'relu', Pad- ding = 'same'
	Dropout	-	-	-	0.5
Transformer Module	Multi-Head Self-Attention	-	-	-	Heads = 8
	Feed-Forward Network	-	-	-	Dense layers
	Positional Encoding	-	-	-	Adds spatial context
Classification Layers	Flatten layer	-	-	-	-
	Dense layer	-	256	-	Activation = 'relu'
	Dropout layer	-	-	-	0.5
	Dense layer	-	4	-	Activation = 'softmax'

Algorithm 1 Trans ResLadderNet for alzheimer's classification

Require: Input shape, num classes, data path, batch size, epochs
Ensure: Trained Trans ResLadderNet model

- 1: **Define ResLadderNet Model**
- 2: $\text{resnet} \leftarrow \text{ResNet50}(\text{weights}='imagenet', \text{include top}=\text{False})$
- 3: **for** each layer $\in$ ResNet50 **do**
- 4: $\text{layer.trainable} \leftarrow \text{False}$
- 5: **end for**
- 6: $x_{\text{in}} \leftarrow \text{Input}(\text{shape} = \text{input shape})$
- 7: $x \leftarrow \text{resnet}(x_{\text{in}})$ ▷ Encoder Path
- 8: $x \leftarrow \text{Conv_2D}(512, (3, 3), \text{activation}='Relu', \text{padding}='Same')(x)$
- 9: $x \leftarrow \text{MaxPooling_2D}(\text{pool size}=(2, 2))(x)$
- 10: $x \leftarrow \text{Dropout}(0.5)(x)$ ▷ Decoder Path
- 11: $x \leftarrow \text{UpSampling2D}(\text{size}=(2, 2))(x)$
- 12: $x \leftarrow \text{Conv_2D}(512, (3, 3), \text{activation}='Relu', \text{padding}='Same')(x)$
- 13: $x \leftarrow \text{Dropout}(0.5)(x)$ ▷ Classification Head
- 14: $x \leftarrow \text{Flatten}()(x)$
- 15: $x \leftarrow \text{Dense}(256, \text{activation}='Relu')(x)$
- 16: $x \leftarrow \text{Dropout}(0.5)(x)$
- 17: $y \leftarrow \text{Dense}(\text{num classes}, \text{activation}='softmax')(x)$
- 18: $\text{model} \leftarrow \text{Model}(x_{\text{in}}, y)$
- 19: $\text{model.compile}(\text{optimizer}='adam', \text{loss}='categorical_crossentropy', \text{metrics} =$
['accuracy']) ▷ Data Preparation
- 20: $\text{datagen} \leftarrow \text{Image Data Generator}(\text{rescale}=1./255, \text{validation split}=0.2)$
- 21: $\text{train gen} \leftarrow \text{datagen.flow from directory}(\text{data path}, \text{target size}=(224, 224), \text{batch}$
size=batch size, class mode) ▷ ViT Integration Block
- 22: $\text{model.fit}(\text{train gen}, \text{steps per epoch}=\lfloor \text{train gen} \rfloor \text{ batch size } \rfloor, \text{epochs}=\text{epochs})$
- 23: $\text{loss, acc} \leftarrow \text{model.evaluate}(\text{validation generator})$
- 24: **print** "Validation Performance: "
- 25: **Classify** Alzheimer Disease

Our proposed hybrid model for AD classification integrates DL techniques to enhance accuracy and robustness. ResNet50 serves as the feature extractor, capturing essential spatial details from medical images. The extracted features are then processed by LadderNet, which refines and reconstructs the information using its dual encode and decode structure. To enhance the model's effectiveness, a Transformer [23] module is integrated after LadderNet, utilizing self-attention techniques to capture extended dependencies and contextual associations. As shown in below Fig. 6.

While ResNet50 and LadderNet handle spatial feature extraction and hierarchical reconstruction, the Transformer module is applied to LadderNet's output feature

maps as a sequence of flattened patches. We add positional encodings to preserve spatial relationships, and then apply multi-head self-attention to capture long-range dependencies across the whole image. The attention outputs are passed through position-wise feed-forward layers, enabling the model to integrate global contextual associations beyond local CNN representations.

The diagram shows how the Transformer module is integrated into the classification framework. After LadderNet processes the image, the resulting feature maps are split into fixed-size patches. These patches are flattened and projected into embeddings, to which positional information is added. The embedded sequence is passed through a Transformer Encoder, consisting of layers with multi-head attention,

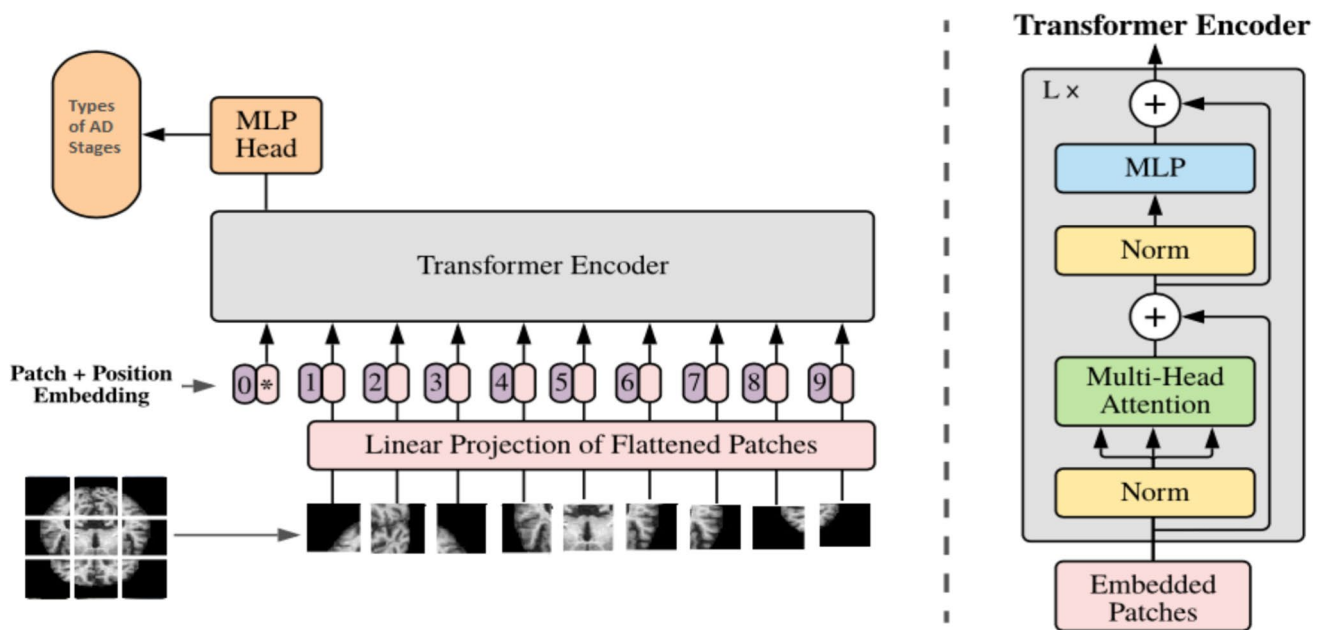


Fig. 6 Architecture of transformer after decoding

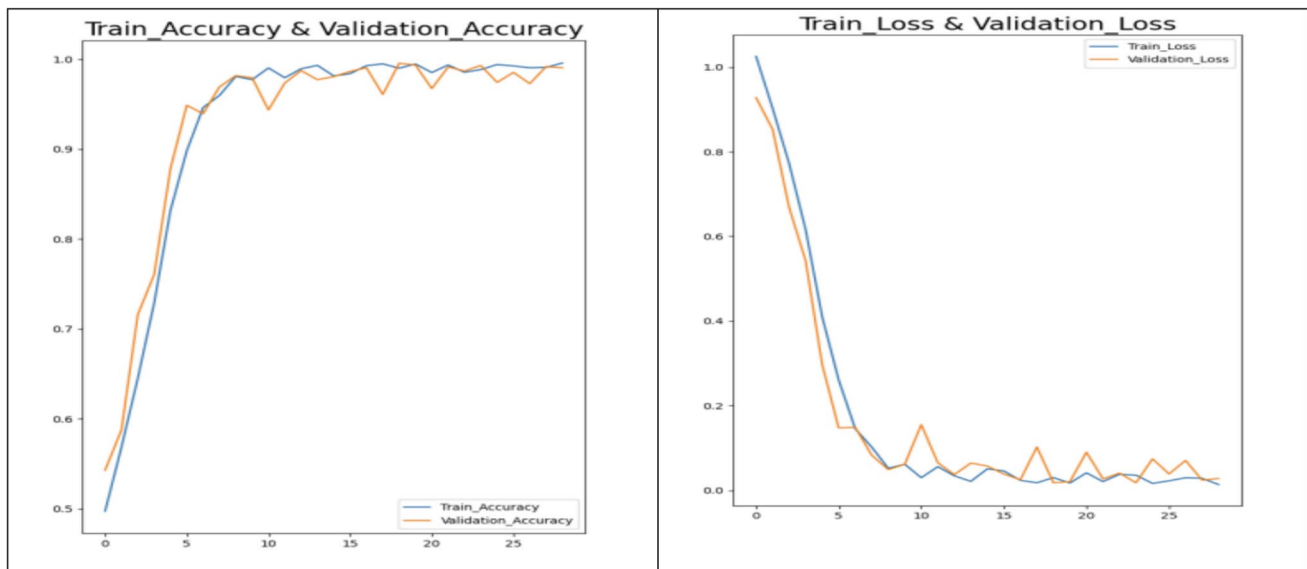


Fig. 7 Graphical presentation of Training accuracy and losses

normalization, and feed-forward networks. Finally, a multi-layer perceptron (MLP) head uses these encoded features to classify the stages of Alzheimer's Disease.

The Transformer enhances [24] feature representation by focusing on the most critical patterns, reducing redundant information, and improving differentiation between various stages of Alzheimer's. This synergy between CNN-based feature extractions, LadderNet's hierarchical processing, and the Transformer's global attention enables precise

classification, improved generalization, and greater interpretability in Alzheimer's disease detection.

Performance Evaluation

The preprocessed data is divided into three parts: 70% for training, 20% for testing, and 10% for validation (Fig. 7). The proposed model is then trained and evaluated against ResNet50

Table 2 Values that collect Epoch time w.r.t. threshold value

Epoch Values	GPU Memory	Boxes Loss	Classes Loss	DFL Losses	Iteration/ Instances	Size of Image	Training Time	Best mAP50	Best mAP50-95
1/100	10G	2.5	2.2	2.2	10	640	01:30	0.82	0.73
10/100	10G	2.4	1.8	2	10	640	01:28	0.84	0.74
20/100	10G	2.3	1.4	1.8	9	640	01:27	0.85	0.74
30/100	10G	2.2	1.2	1.7	9	640	01:26	0.86	0.77
40/100	10G	2.1	1	1.6	8	640	01:26	0.86	0.78
50/100	10G	2	0.8	1.5	8	640	01:22	0.88	0.81
60/100	10G	1.9	0.6	1.4	7	640	01:19	0.91	0.83
70/100	10G	1.8	0.4	1.3	7	640	01:19	0.93	0.83
80/100	10G	1.7	0.4	1.2	6	640	01:18	0.94	0.84
90/100	10G	1.6	0.2	1.1	5	640	01:18	0.96	0.86
100/100	10G	1.5	0.2	2.2	4	640	01:30	0.97	0.87

Table 3 Training accuracy

Algorithm	Precision	Recall	F1-Score	Loss	MCC
ResNet50	0.89	0.88	0.89	0.21	0.86
LadderNet	0.93	0.92	0.93	0.15	0.91
ResLadderNet	0.95	0.94	0.95	0.11	0.93
Transformer ResLadderNet	0.98	0.97	0.98	0.06	0.97

and LadderNet. The obtained values for accuracy, F1 score, precision, and recall are presented in the table below (Table 2).

We trained our hybrid model, combining ResNet50 and LadderNet (ResLadderNet), and compared its performance

with individual models—ResNet50 and LadderNet—to assess their effectiveness in Alzheimer's disease classification. The training comparison finding matrices store in below Table 3, graphically shown in Fig. 8 and testing comparison shown in Table 4 and graphically shown in Fig. 9.

After establishing the baseline performance, we integrated a Transformer module to enhance feature propagation and contextual learning, leveraging its attention mechanism for capturing long-distance dependencies. The table below presents a comparison of ResLadderNet with other models, demonstrating its superior performance.

To validate the robustness of our proposed model, we conducted testing on three distinct datasets: Harvard,

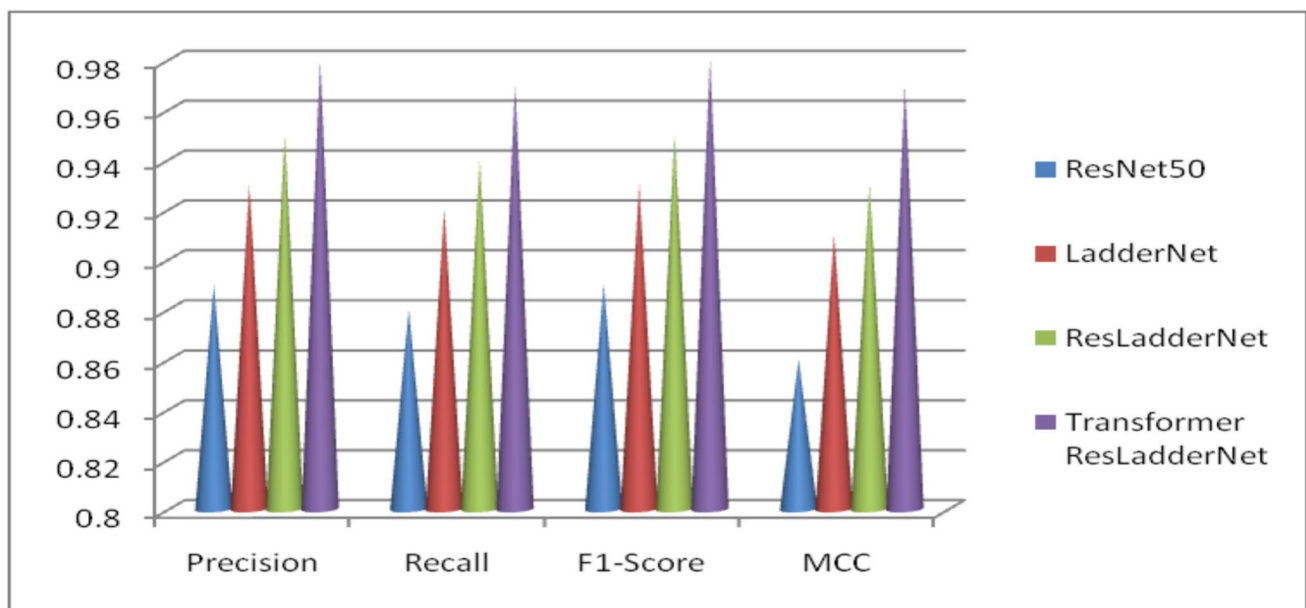
**Fig. 8** Graphical presentation of training accuracy

Table 4 Testing accuracy

Algorithm	Precision	Recall	F1-Score	Loss	MCC
ResNet50	0.85	0.83	0.86	0.28	0.81
LadderNet	0.88	0.89	0.89	0.22	0.86
ResLadderNet	0.92	0.91	0.91	0.16	0.89
Transformer ResLadderNet	0.97	0.93	0.95	0.08	0.95

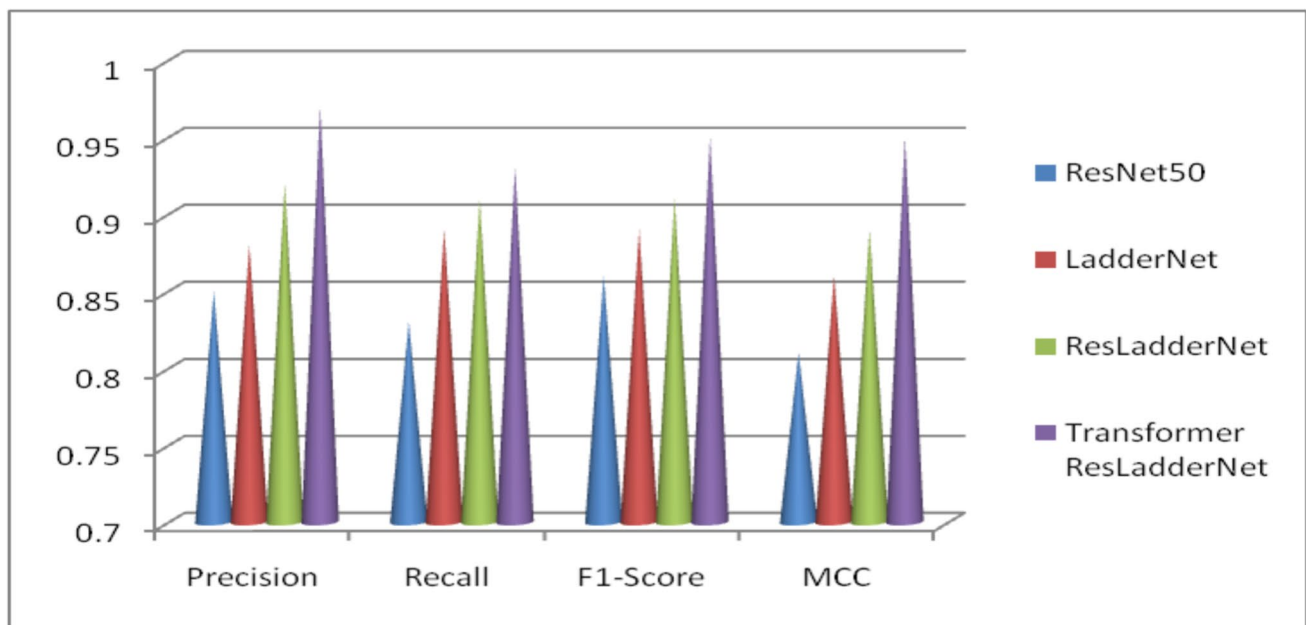
ADNI, and Kaggle. The model consistently achieved high accuracy, precision, recall, and F1-scores, demonstrating strong generalization across diverse data sources. The performance over different dataset are given in below Table 5 and graphically in Fig. 10.

To evaluate the matrices of the proposed work, we implemented K-Fold cross validation with twofold, fivefold, and sevenfold splits. The twofold validation divides the dataset into two equal parts for training and testing, providing a rapid assessment but with a higher chance of variance due to limit training data per fold. The fivefold validation offers a more balanced approach, improving stability while maintaining computational efficiency.

Meanwhile, the sevenfold validation further enhances generalization by ensuring that each data point is used in both training and testing multiple times, reducing bias and mitigating overfitting. Using multiple validation folds allows for a more comprehensive assessment of the model's reliability and effectiveness across different training-test splits. Finding of each fold is store in Table 6 and graphically present in Fig. 11.

To enhance model transparency and interpretability, we employed Gradient-weighted Class Activation Mapping (Grad-CAM) on the final convolutional layers of the ResNet50 backbone. This technique visualizes the key regions in input images that most strongly influenced the model's classification decisions across the four Alzheimer's Disease (AD) progression stages. The resulting activation maps help validate the model's focus on clinically relevant areas. A representative Grad-CAM visualization is shown in Fig. 12 below.

We also store the confusion matrix to assess the model's performance across various classes. This allows us to analyze misclassifications and understand the precision and recall for each category, ensuring a thorough assessment of

**Fig. 9** Graphical presentation of testing accuracy**Table 5** Testing evaluation across multiple datasets

Dataset	Accuracy (%)	Precision (%)	Sensitivity (%)	Specificity (%)	F1-Score (%)
Harvard	97.2	97	93	91	95
ADNI	98.92	98.78	95.95	92.4	93.18
Kaggle	98.93	98.79	95.98	93.21	98.93

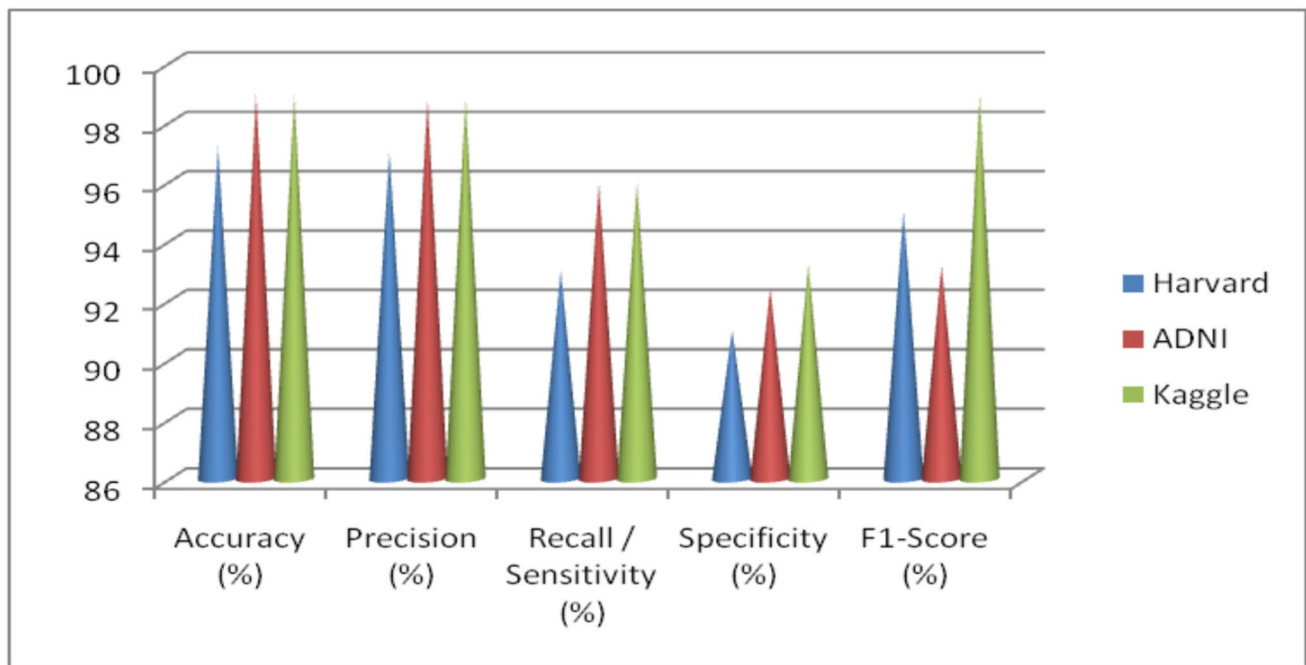


Fig. 10 Graphical presentation of different dataset based performance

Table 6 K-fold testing accuracy

K-Fold Validation	Accuracy in %	Precision in %	Recall in %	F1-Scores in %
2-Fold	98.85	98.7	95.8	98.1
5-Fold	98.9	98.75	95.9	98.2
7-Fold	98.92	98.78	95.95	98.38

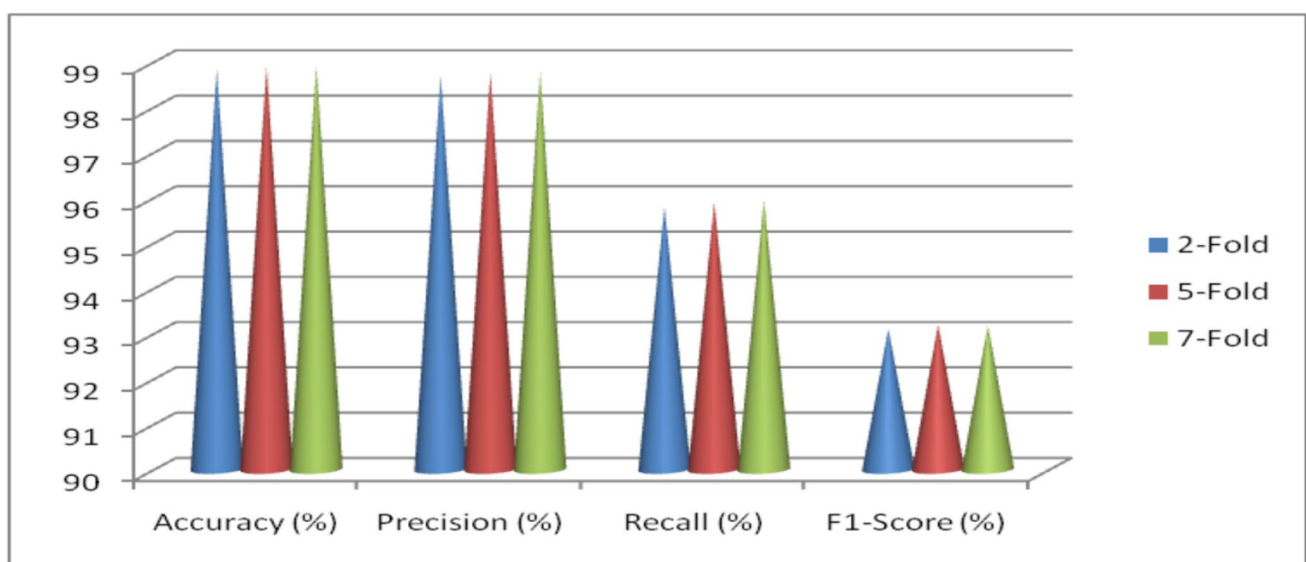


Fig. 11 Graphical presentation of K-fold findings

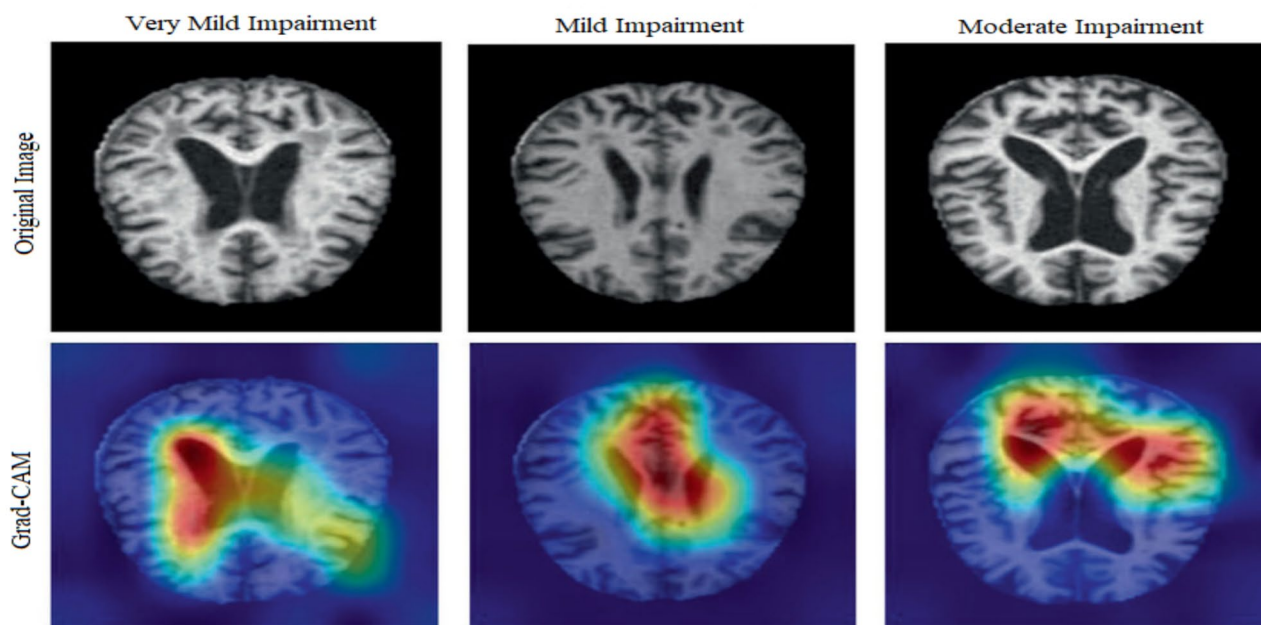


Fig. 12 Visualization of Grad-CAM over original image

Fig. 13 Confusion matrix of proposed model

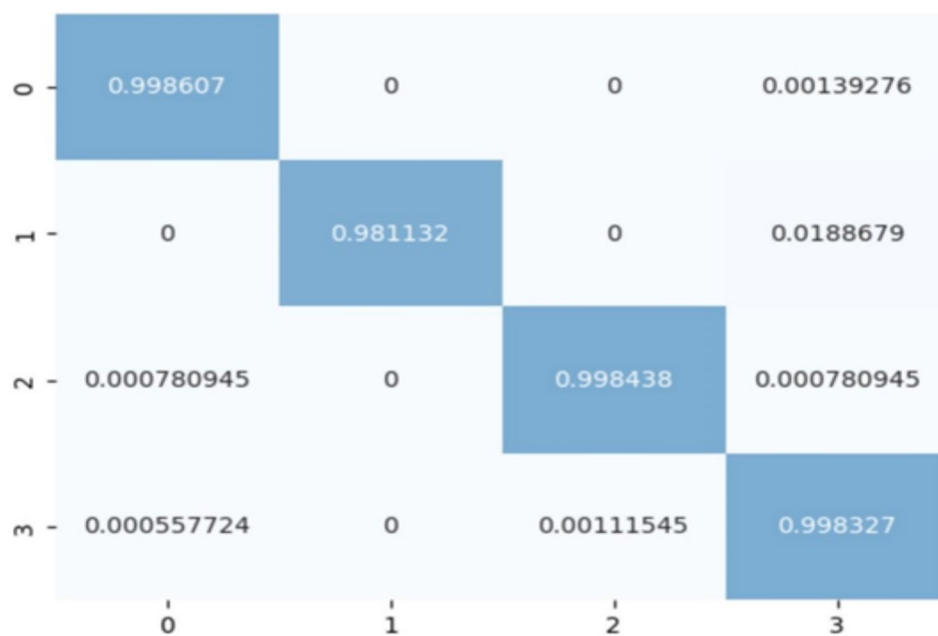


Table 7 Comparison of proposed model with existing Model

References	Year	Technique	Accuracy	F1 Score	Sensitivity	Specificity
[2]	2024	MCLSO-SpinalNet	92.6%	91.7%	93.6%	93.3%
[4]	2021	CNN based	97.5%	95.01%	95.60%	95.3%
[5]	2023	EfficientNetB	83%			
[6]	2022	Resnet50	83.8%	81.9%		
		DenseNet 169	97.7%	88.7%		
[8]	2023	SVM & Resnet 50	98.71%			
[9]	2023	DenseNet-AM	91.6%		93.9%	89.6%
[10]	2023	ConvNeXt	78.95%	77.08%	80.91%	76.60%
[11]	2021	DNN	76.93	74.93%	76.45%	72.73%
Proposed Model			98.93%	98.79%	95.98%	93.21%

the model's accuracy in detecting AD stages below Fig. 13 shows the graphical presentation of confusion matrix.

We also compare our proposed work with other researchers proposed model and comparison table is given in below Table 7.

Conclusion and future work

Our study introduces Trans ResLadderNet, achieving exceptional performance in Alzheimer's disease (AD) classification with an accuracy of 98.93%, F1 score of 98.79%, sensitivity of 95.98%, and specificity of 93.21%. This innovative approach not only enables early detection and personalized treatment planning but also deepens insights into AD pathology through robust feature extraction and hierarchical learning. Future endeavors will focus on enhancing model generalizability across diverse patient demographics and integrating multi-modal data for more nuanced disease characterization. These advancements aim to further refine diagnostic accuracy, support timely interventions, and drive forward therapeutic advancements, ultimately improving outcomes for AD patients worldwide.

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Declarations

Ethics Approval and Consent to Participate Not Applicable.

Consent for Publication All coauthors consented for the publication.

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